Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: Bachelor of Science in Civil Engineering Major in Structural Engineering

Primary Job Roles students are prepared for in the program	**Design structures such as buildings, bridges and tunnels; prepare structural reports, design and drawings; provide technical advice to architects and contractors; Monitor and inspect structures during and after construction; examine existing structures and provies advise on how to improve their structural integrity.**
Secondary Job Roles students are prepared for in the program	**Consultation, Design, Preparation of plans, specifications and estimates; Construction, Installation and Supervision of horizontal and vertical structures; Teaching of Professional Civil Engineering Subjects; Assess environmental impact of projects; lead field investigations, research, testing and analysis in solving civil engineering problems.**

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program: CMO 101, Series of 2017

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of policies, standards and guidelines in the establishment of an outcomes-based education (OBE) system in higher education institutions offering engineering programs as advocated under CMO 37 s. 2012 and policy-standard to enhance quality assurance (QA) in Philippine higher education through an outcomes-based and typology-based QA as advocated under CMO 46 s. 2012, and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following policies and standards are hereby adopted and promulgated by the Commission.

This CMO integrates the establishment of an outcomes-based Based on the Guidelines for the Implementation of CMO No. 46 series of 2012 and CMO 37 s. 2012, this PSG implements the shift to outcomes-based education (OBE) leading to competency-based standards. It specifies the "core competencies" expected of Bachelor of Science in Civil Engineering (BSECE) graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description (CMO 101 Series of 2017):

Definition
Civil Engineering is a profession that applies the basic principles of sciences in conjunction with mathematical and computational tools to solve problems associated with developing and sustaining civilized life on our planet. Civil Engineering works are generally one-of-a kind projects; they are often grand in scale; and they usually require cooperation among professionals of many different principles. The completion of a Civil Engineering project involves the solution of technical problems in which information from numerous sources and myriad non-technical factors play a significant role.

Degree Name
The degree program described herein shall be called Bachelor of Science in Civil Engineering (BSCE) major in Structural Engineering

Nature & Scope of the practice of a CE graduate
Civil engineering is one of the broadest engineering disciplines both in terms of the range of problems that fall within its purview and in the range of knowledge required to solve those problems. Some of the most common examples of Civil Engineering works include bridges, buildings, dams, airports and hangars, ports and harbors, highways, and railways, tunnels, river and shore improvements, lighthouses, dry-docks, irrigation, flood protection, drainage, water supply, and towers. Enumeration of any work in this section shall not be construed as excluding any other works requiring civil engineering knowledge and application.

VISION
A PREMIER UNIVERSITY IN THE ASEAN REGION

MISSION
USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.

CORE VALUES
USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

Program Educational Objectives	Mission-Vision and Core Value Alignment						Reference Source or CMO Provision Compliance
	Vision	Mission	Core Value				
			U	S	E	P	

PEO1	Demonstrate expertise in the field of ~~Electronics~~ Civil Engineering	√	√		√	√	√	USEP Brochure, CMO 101
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√		√	USEP Brochure, CMO 101
PEO3	Demonstrate gender sensitive leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√		√	√	USEP Brochure, CMO 101
PEO4	Exhibit globally competitive skills	√	√		√	√	√	USEP Brochure, CMO 101

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Institutional Graduate Attributes:

IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

Institutional Graduate Attribute		Program Outcomes		Performance Indicators
IGA1	a	Apply knowledge of mathematics and sciences to solve engineering problems;	1	Apply concepts in mathematics to solve engineering problems
IGA4			2	Apply principles in chemical sciences to solve engineering problems.
IGA6			3	Apply principles in physical sciences to solve engineering problems
IGA1, IGA4	b	Design and conduct experiments, as well as to analyze and interpret data	1	Perform laboratory experiments as prescribed by standard procedures.
IGA5, IGA6			2	Design an experiment to achieve a specific objective
IGA1, IGA2	c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	1	Develop an engineering solution that have various constraints as factors
IGA3, IGA4			2	Design a system or process based on various constraints
IGA5, IGA6				
IGA2	d	Function on multidisciplinary teams	1	Shows tact in dealing with people he/she comes in contact with.
IGA3			2	Capable of learning from and listening to co-members
IGA6			3	Shows respect and courtesy in dealing with peers and superiors.

IGA1	e	Identify, formulate, and solve engineering problems	1	Interpret statement problems into mathematical expressions.
IGA4			2	Apply concepts, theories, and principles in solving engineering problem(s)
IGA6			3	Solve specific engineering problem related to discipline.
IGA4	f	Understand professional, social and ethical responsibility	1	Internalizes the professional code of ethics for specific discipline.
IGA5			2	Applies the principles of professional ethical standard and social responsibility to actual practice
IGA6				
IGA4	g	Communicate effectively	1	Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5			2	Articulate the convention and style in written communication that conveys complete thought and information.
IGA6			3	Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3	h	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1	Relate engineering solutions to global, economic, environmental, and societal context.
IGA4, IGA6			2	Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2	i	Recognize the need for, and an ability to engage in life-long learning	1	Demonstrate ability to learn independently.
IGA3, IGA4			2	Shows appreciation and understanding of the engineering profession.
IGA5, IGA6			3	Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3	j	Apply knowledge of contemporary issues	1	Identify the socio-economic, political, and environmental issues related to the practice of engineering
IGA4, IGA5			2	Explain the possible impact of technology to local, national, and international issues
IGA6			3	Shows leadership skills in managing a multidisciplinary group.
IGA1	k	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
IGA4			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
IGA6			3	Use modern technology in performing engineering practice.
IGA1, IGA2	l	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
IGA3, IGA4			2	Shows leadership skills in managing a multidisciplinary group.
IGA5, IGA6			3	Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

List courses such that first year first semester courses appear first, until the last semester of the last year of the degree program. Exclude in the list the following: (a) General Education courses (b) Elective courses. As much as possible, observe the following:

a. *For every course of 3-units, do not exceed 3-4 Graduate Outcomes you will be focusing and targeting to develop in the course*
b. *For every course of 1-units, do not exceed 1-2 Graduate Outcomes you will be focusing and targeting to develop in the course*
c. *For every course of 5-units, do not exceed 4-5 Graduate Outcomes you will be focusing and targeting to develop in the course*
d. *Ensure that all the Graduate Outcomes have courses developing it for every year level*
e. *Total the number of X's horizontally per course and vertically per Graduate Outcome*

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Program Outcomes	
Upon completion of the program, the University of Southeastern Philippines Civil Engineering students will be able to:	
a	Apply knowledge of mathematics and sciences to solve Civil Engineering problems;
b	Design and conduct experiments in order to understand a phenomenon, prove an idea, test an engineering solution and to make sound scientific conclusions , as well as to analyse and interpret data using various models and frameworks;
c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards
d	Function on multidisciplinary and multi-cultural teams
e	Identify, formulate, and solve Civil Engineering problems
f	Understand professional, social and ethical responsibility
g	Communicate effectively through writing, listening, presenting, observing, questioning, focusing, and connecting on complex Civil Engineering activities with various engineering communities including engineering experts and society at large
h	Understand the impact of Civil Engineering solutions in a global, economic, environmental, & societal context through broad education
i	Recognize the need for, and an ability to engage in life-long learning to keep abreast of emerging technologies and continually learn new knowledge
j	Know contemporary issues and the ensuing responsibilities relevant to professional engineering practice
k	Use techniques, skills, and modern engineering tools necessary for Civil Engineering practice.
l	know and understand of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments
m	understand at least one specialized field of civil engineering practice

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes												
				a	b	c	d	e	f	g	h	i	j	k	l	m
1	1	3	Purposive Communication							D						
		3	Mathematics in the Modern World	I				I								
		3	The life and Works of Jose Rizal				I		I			I				
		3	Engineering Calculus 1	I												
		4	Chemistry for Engineers	I	I									I		
		1	Engineering Drawing and Plans				E		I	I				E		
		2	Civil Engineering Orientation						I		I		I			
		2	Movement Enhancement				I		I			I	I			
		3	National Service Training Program 1				I		I			I	I			
	TOTAL	**24**														
	2	3	Understanding the Self						I			I			I	
		3	Ethics						I						I	
		3	Science, Technology and Society				I			I	I	I				
		3	Engineering Calculus 2	I												
		4	Physics for Engineers	I	I									I		
		1	Computer Aided Design and Drafting	I										D		
		2	Engineering Geology					E	E		D	E	E	I		
		2	Fitness Exercise				I		I			I	I			
		3	National Service Training Program 2				I		I			I	I			
	TOTAL	**24**														
OFF SEMESTER 1		3	Computer Fundamentals and Programming	I										E		
	TOTAL	**3**														
2	1	3	Readings in the Philippine History						I				I			
		3	Popular Literature				I				I		I			
		3	Contemporary World								I	I	I			
		3	Differential Equations	I										I		
		3	Engineering Data Analysis	I	I									I		
		5	Statics and Dynamics of Rigid Bodies	I												
		3	Engineering Economics	E										E		

		2	Physical Activities Towards Health & Fitness I				I		I			I	I				
	TOTAL	**25**															
		3	Arts Appreciation			I	I										
		3	Numerical Solutions to CE Problems	E				D		E		E		E			
		3	GE Elective 1														
		4	Mechanics of Deformable Bodies	E				D									
	2	3	Methods of Research						E	E	E		E		E		
		3	Fundamentals of Surveying		D		E	I		E				E			
		2	Fundamentals of Surveying, Field		D		E	I		E				E			
		3	Building System Design			E	E	E	I	I	I		I	E			
		2	Physical Activities Towards Health & Fitness II				I		I			I	I				
	TOTAL	**26**															
OFF SEMESTER 2		3	Engineering Utilities 1			E	E		E	E		E					
		3	Engineering Utilities 2			E	E		E	E		E					
	TOTAL	**6**															
		3	GE Elective 2														
		3	Engineering Management							E		E			E		
		4	Structural Theory	E	I			E						I			
		3	Principle of Transportation Engineering	E					D								
	1	3	Geotechnical Engineering, Lecture	E	D	D	E	E		E			E	E			
		1	Geotechnical Engineering, Lab	E	D	D	E	E		E			E	E			
		2	Quantity Surveying	E			E		E	E		E		E	E		
		4	Hydraulics, lecture	E	D		E	E		E							
3		1	Hydraulics, lab	E	D		E	E		E							
	TOTAL	**24**															
		3	Highway and Railroad Enginering	E		E		E		E							
		2	Construction Materials and Testing, lecture	E	D		E			E			E	E			
		1	Construction Materials and Testing, lab	E	D		E			E			E	E			
	2	4	Principle of Steel & Timber Design	E	I									E			
		4	Principle of Reinforced / Prestressed Concrete	E	I									E			
		2	Hydrology	E				E		E				E			
		3	Construction Methods and Project Management	E				E	E	D				D	E		

		2	CE Laws, Ethics and Contracts						D	E							
		3	Technoprenuership					E	E		D	D	D	D			
	TOTAL	**24**															
OFF SEMESTER 3		1	Basic Driving and Automotive				D		D								
		1	Scaffold Erection	D		D	D			D	D			D			
		1	Disaster Risk Management	E		E	E		E		E		E				E
	TOTAL	**3**															
4		3	CE Project (Specialization Project, Thesis)*	E	D	D	D	D	D	D	D			D	D		
		1	Field trip and Seminars	E		E	E		E		E		E				E
		3	Professional Course - Specialized 1	E		E		E		E			E	E			D
	1	3	Professional Course - Specialized 2	E		E		E		E			E	E			D
		3	Professional Course - Specialized 3	E		E		E		E			E	E			D
		3	Professional Course - Specialized 4	E		E		E		E			E	E			D
		3	Professional Course - Specialized 5	E		E		E		E			E	E			D
	TOTAL	**19**															
		3	CE Project (Specialization Project, Thesis)*	E	D	D	D	D	D	D	D			D	D		
	2	2	Correlation Course	E	D	D	E	E		E			E	E			
		2	Specialization On-the-Job Training (320 hours)	E	D	D	D	D	D	D	D			D	D		
		1	FREE ELECTIVE	E		E	E		E		E		E				E
	TOTAL	**8**	**STRUCTURAL ENGINEERING TRACK**														
4		3	Computer Softwares in structural Analysis	E		E		E		E			E	E			D
		3	Earthquake Engineering	E		E		E		E			E	E			D
		3	Design of Steel Structures	E		E		E		E			E	E			D
	1	3	Reinforced Concrete Design	E		E		E		E			E	E			D
		3	Prestressed Concrete Design	E		E		E		E			E	E			D
		3	Bridge Engineering	E		E		E		E			E	E			D
		3	Foundation and Retaining Wall Design	E		E		E		E			E	E			D

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

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